



Nazmi Yücel Çan

MACHINE LEARNING ENGINEER • Giresun, Türkiye (Open to Relocation / Remote)

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TECHNICAL PROFILE

Machine Learning Engineer specializing in model development, optimization, and edge AI deployment. Reduced mT5 SLM model size 5x (from 500MB to 100MB) via pruning, 8-bit LoRA, and INT8 quantization. Following computer vision experience, currently focusing on Natural Language Processing (NLP). Co-authored a research paper published in El-Cezeri Journal (2026). I developed Zettelkasten AI Notes, a privacy-first desktop knowledge management system powered by Vulkan-accelerated local GGUF models and Google Gemini, ranking in the top 10 finalists of the Pupilica AI Hackathon.

SKILLS

Programming Languages

Python (Advanced), SQL, Bash/Shell Scripting

ML & AI Frameworks

TensorFlow, PyTorch, Keras, Scikit-learn,
Hugging Face (LoRA/PEFT), ONNX Runtime,
llama-cpp-python

Core Domains

Computer Vision (OpenCV, YOLOv8), NLP
(Transformers), Data Preprocessing &
Evaluation, Model Optimization (Quantization &
Pruning), MLOps Fundamentals

Tools & Platforms

Git/GitHub, Linux (Ubuntu/Debian), PyQt5, Flet, Google Gemini API

Languages: Turkish (Native), English (Upper-Intermediate)

EXPERIENCE

Machine Learning Intern at Deep Think Technology

02/2026 – 06/2026

- Developed the industrial "Apple Detection and Quality Analysis System", integrating real-time YOLOv8-seg fruit detection, CIEDE2000 color quality analysis, physical dimension measurement, and camera SDK into a unified PyQt5 desktop application.
- Trained and deployed computer vision models on embedded devices for real-time edge AI inference.
- Implemented data preprocessing and model evaluation workflows for edge computing hardware, addressing real-world latency and resource constraints.

PROJECTS

mT5 Bidirectional EN-TR Machine Translation (ONNX INT8 & ORT Mobile)

Python, PyTorch, Hugging Face PEFT (LoRA), ONNX Runtime, INT8 Quantization, mT5 SLM

- Fine-tuned an mT5-small model with 8-bit LoRA for bidirectional English-Turkish translation, achieving a 0.6245 METEOR score. Reduced model size 5x (from 500MB to 100MB) and optimized performance for resource-constrained environments by applying model pruning, ONNX compilation, and INT8 quantization for mobile deployment via ONNX Runtime (ORT Mobile).

Kaggle

Zettelkasten AI Notes (Pupilica AI Hackathon — Top 10 Finalist)

Python, Flet, SQLite, llama-cpp-python (Vulkan GPU), Google Gemini AI

- A desktop knowledge management application built around the Zettelkasten method using Python and Flet. Features a Dual AI Engine providing both cloud (Google Gemini) and 100% offline local inference (GGUF models accelerated via Vulkan GPU) to extract atomic notes and semantic links from PDFs. Includes an in-app model downloader, hardware resource guard, live Markdown editor with LaTeX math, smart bidirectional WikiLinks, and an interactive canvas mind map. All data is persisted in a local SQLite database with WAL mode.

GitHub

Apple Detection and Quality Analysis System

Python, PyQt5, OpenCV, YOLOv8, MVS SDK

- An industrial computer vision application featuring real-time object detection, physical dimension measurement, and CIEDE2000 color quality analysis. Focused on system integration, designing the application architecture, and orchestrating the end-to-end pipeline—integrating the camera SDK, real-time YOLOv8-seg model inference, and measurement modules into a unified desktop interface.

[GitHub](#)

Arduino PC System Monitor

Python, Arduino (C++), PlatformIO, TinyML (SVM), scikit-learn

- A TinyML-powered system that monitors PC hardware resources (CPU, RAM, GPU, VRAM) and automatically optimizes power consumption. The system extracts hardware telemetry in real-time, executing an on-device SVM classification model on an Arduino Leonardo to determine and adjust the system's power state.

[GitHub](#)

PUBLICATIONS

Deep Learning Models Integrating Attention Mechanisms For Military Camouflaged Object Detection — Nilgün Şengöz, Gül Karaman, Mert Samet Çeliker, Nazmi Yücel Çan

El-Cezeri Fen ve Mühendislik Dergisi (2026) — <https://doi.org/10.31202/ecjse.1747013>

EDUCATION

Bachelor of Science in Information Systems and Technologies

09/2022 – 06/2026

Burdur Mehmet Akif Ersoy University

GPA: 3.17/4.00

REFERENCES

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